



CAS PLATFORM

Operator Portal User Guide

Conjunction Decision Support for LEO Operators

RELEASE

April 2026

MATURITY

TRL 5

AUDIENCE

LEO Satellite Operators

ENGINE

Decision Engine v0.7

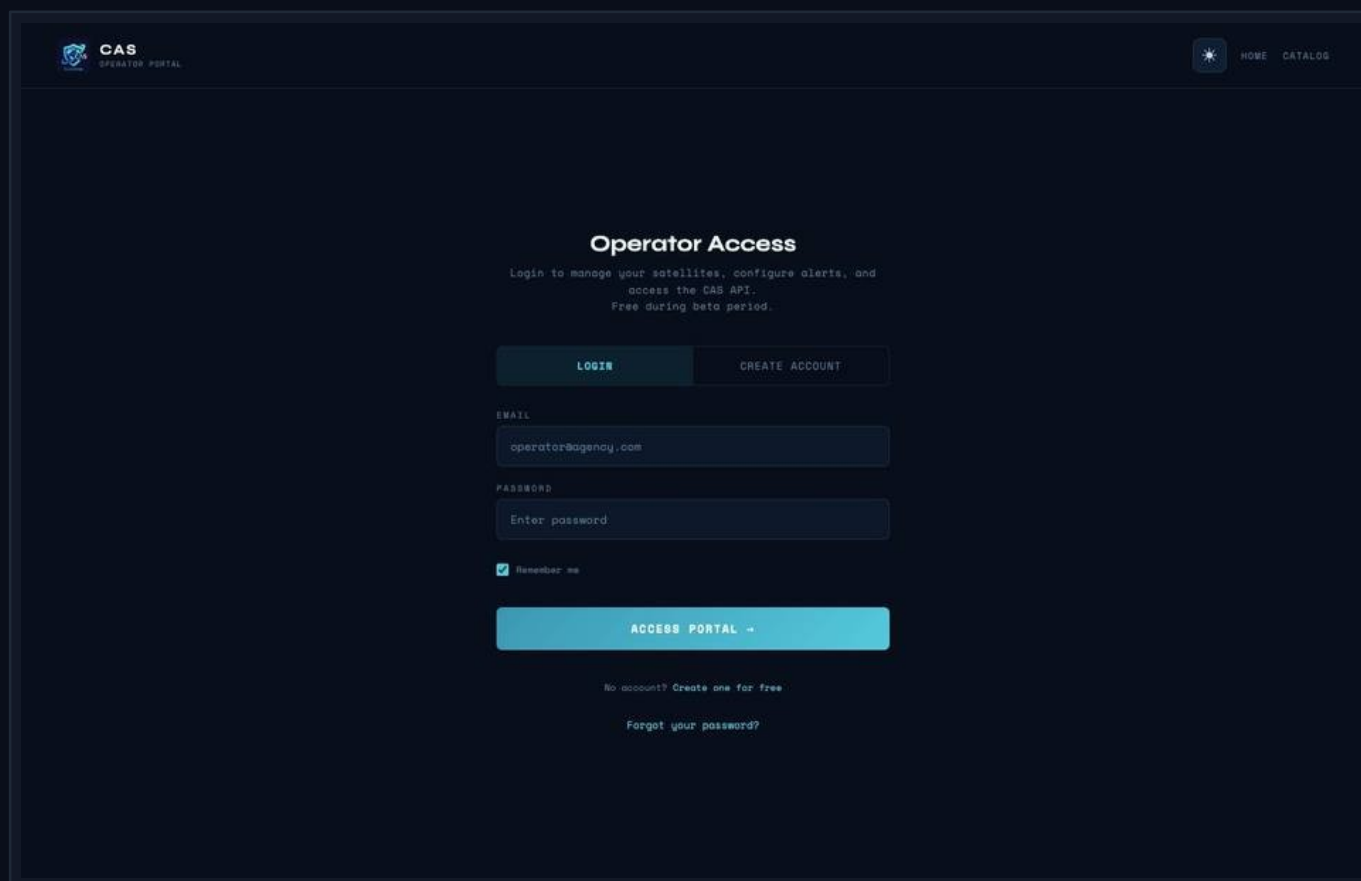
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What this guide covers

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Operator Access

Email verification, JWT auth, secure session handling



The screenshot shows the 'Operator Access' login page. At the top left is the 'CAS OPERATOR PORTAL' logo, and at the top right are links for 'HOME' and 'CATALOG'. The main heading is 'Operator Access', followed by a subtext: 'Login to manage your satellites, configure alerts, and access the CAS API. Free during beta period.' Below this are two buttons: 'LOGIN' and 'CREATE ACCOUNT'. The 'LOGIN' button is highlighted. Underneath are input fields for 'EMAIL' (containing 'operator@agency.com') and 'PASSWORD' (with a placeholder 'Enter password'). A 'Remember me' checkbox is checked. A large blue button labeled 'ACCESS PORTAL' is below the inputs. At the bottom, there are links for 'No account? Create one for free' and 'Forgot your password?'.

WHAT IT SHOWS

Entry point for the operator portal. Login or create an account. Verification gates portal features behind a confirmed email.

HOW TO USE

- Enter operator email and password
- Toggle Remember me for persistent session
- First-time users: Create Account, then click verification link
- Forgotten password resets via email link

ENGINE BEHAVIOR

SHA256-hashed credentials, JWT issued on success. Verification email signed with SPF/DKIM for deliverability.

OPERATIONAL VALUE

Beta access is free for qualifying operators. Verification ensures alert emails reach an inbox the operator actively monitors.

Overview Dashboard

Status, satellites, priority event, recent CDM feed

HOME

CATALOG

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Overview

System status and recent activity

ENGINE STATUS

ONLINE

CAS v0.7 - Decision Engine

MY SATELLITES

15

In your watchlist

RED ALERTS

177

Active high-risk events

YOUR TIER

ENTERPRISE

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SPACE WEATHER

ELEVATED

Kp: 5.0 - P10: 7.143 sfs - X-ray: X2.5

YOUR SATELLITES

CONNECTA IOT-13 (87490)

CONNECTA IOT-10 (87390)

CONNECTA IOT-11 (84587)

CONNECTA IOT-10 (84586)

CONNECTA IOT-12 (84563)

CONNECTA IOT-9 (84534)

CONNECTA IOT-7 (82715)

CONNECTA IOT-8 (82700)

CONNECTA IOT-5 (82703)

CONNECTA IOT-8 (82696)

CONNECTA IOT-4 (80524)

CONNECTA IOT-2 (80522)

CONNECTA IOT-3 (80476)

CONNECTA T1.2 (66012)

CONNECTA IOT-1 (80472)

VIEW ALL

HIGHEST PRIORITY EVENT

CONNECTA IOT-5

execute

RED - Urgent maneuver recommended. Post-7h-8 (~17h+ RED threshold). Cascade engine identified 1 secondary risk if no action; proposed retrograde maneuver eliminates it. Lead time: 9.2h. Fuel cost: ~25g for 150kg eat class.

Pd: 1.70h Miss: 140m Time: 8h 11m Alerts: 12 - 3 - 7

VIEW ALL DECISIONS

Recent conjunctions

Latest events from Space-Track CDM feed

RISK	OBJECT 1	OBJECT 2	MISS (M)	PC	TCA
RED	CZ-BA DEB	FENGYUN 1C DEB	8	1 in 175	2026-04-24T14:07
RED	FENGYUN 1C DEB	CZ-BA DEB	2	1 in 134	2026-04-24T14:07

WHAT IT SHOWS

Single-screen situational awareness: engine status, watchlist size, RED alert count, account tier and current space weather, plus the highest-priority decision card.

HOW TO USE

- Scan the five status cards at the top of the page
- Open the highlighted priority event for full context
- Jump into the watchlist via the satellite chips
- Triage incoming CDMs from the Recent conjunctions table

ENGINE BEHAVIOR

Each card pulls from a dedicated service: Decision Engine v0.7 (priority event), watchlist scanner, Space-Track CDM feed, NOAA SWPC indices.

OPERATIONAL VALUE

Replaces the daily check-in across multiple feeds with one page. The priority card surfaces what needs human attention right now.

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Watchlist Management

Add satellites, view altitudes, trigger on-demand scans



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Reentries

Add your satellites and monitor conjunction risks in real time

ADD SATELLITE TO WATCHLIST

Search by name (e.g. ISS, STARLINK, CONNECTA)...

SCAN NOW

REFRESH

Your watchlist

NORAD ID	NAME	ALTITUDE	LAST SCAN	ACTION
67400	CONNECTA IOT-13	514.1 km	25.04.2026 08:48:22	REMOVE
67390	CONNECTA IOT-16	513.7 km	25.04.2026 08:48:23	REMOVE
64557	CONNECTA IOT-11	504.2 km	25.04.2026 08:48:24	REMOVE
64555	CONNECTA IOT-10	505.5 km	25.04.2026 08:48:25	REMOVE
64553	CONNECTA IOT-12	503.9 km	25.04.2026 08:48:26	REMOVE
64534	CONNECTA IOT-9	505.6 km	25.04.2026 08:48:27	REMOVE
62715	CONNECTA IOT-7	585.3 km	25.04.2026 08:48:27	REMOVE
62708	CONNECTA IOT-6	585.2 km	25.04.2026 08:48:28	REMOVE
62703	CONNECTA IOT-5	583.6 km	25.04.2026 08:48:29	REMOVE
62695	CONNECTA IOT-8	588.3 km	25.04.2026 08:48:30	REMOVE
60524	CONNECTA IOT-4	445.6 km	25.04.2026 08:48:31	REMOVE
60522	CONNECTA IOT-2	495.3 km	25.04.2026 08:48:31	REMOVE
60475	CONNECTA IOT-3	496.4 km	25.04.2026 08:48:32	REMOVE
58012	CONNECTA T1.2	424.8 km	25.04.2026 08:48:33	REMOVE

WHAT IT SHOWS

The operator's watchlist of monitored satellites with NORAD ID, name, altitude and last-scan timestamp. Hourly auto-scan plus on-demand Scan Now.

HOW TO USE

- Search by name (e.g. CONNECTA) and add to watchlist
- Scan Now to force an immediate conjunction screen
- Refresh to update altitude and last-scan columns
- Remove satellites no longer in scope

ENGINE BEHAVIOR

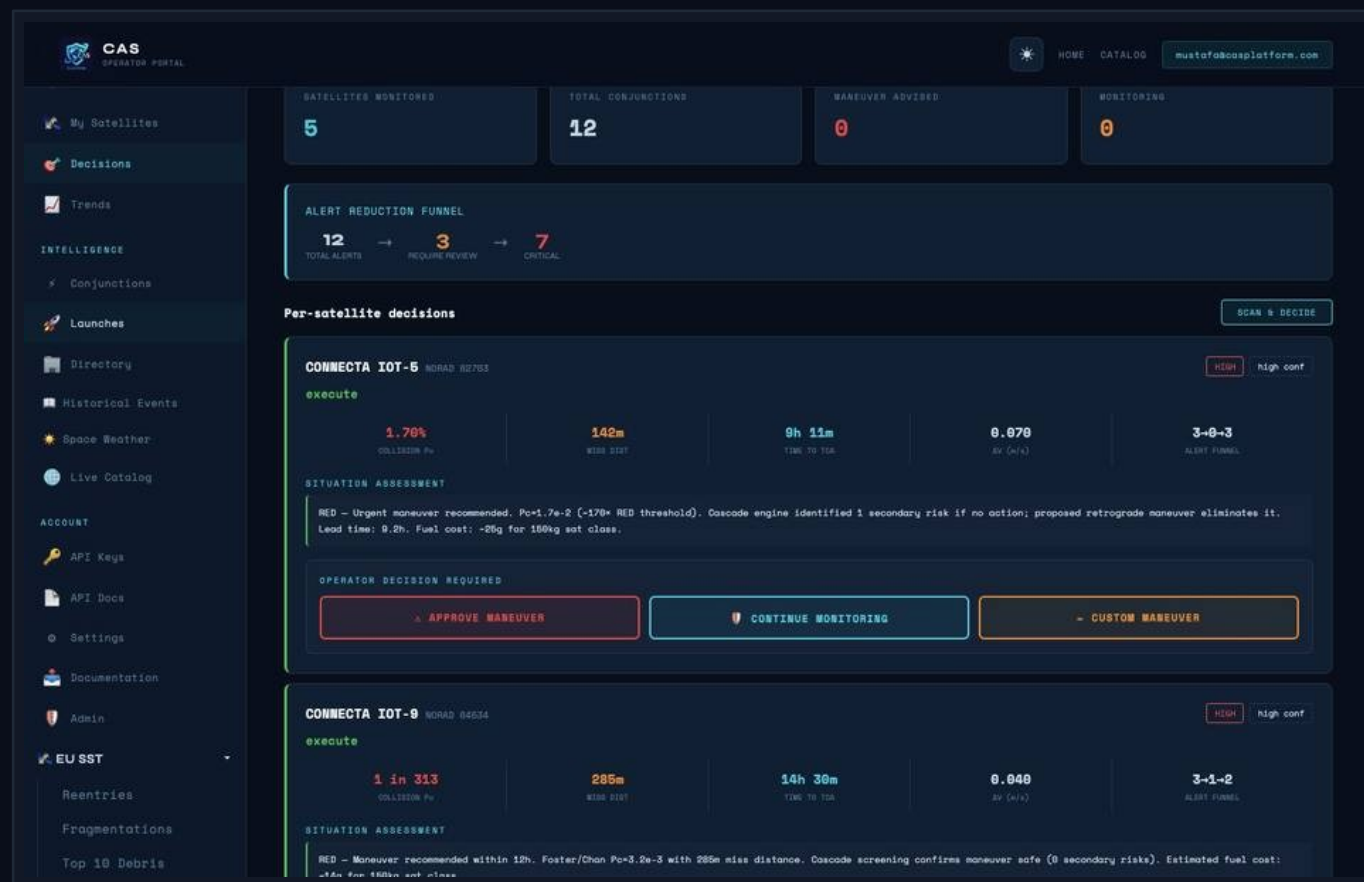
Hourly background scanner correlates each watchlist NORAD ID against incoming CDMs and the catalog. Per-tier maximum satellite limits enforced server-side.

OPERATIONAL VALUE

Operators monitor only what they own. Per-satellite trend, decision and alert pipelines key off this list — no irrelevant noise from the broader catalog.

Per-Satellite Decision Support

Risk-classified events with maneuver guidance and cascade analysis



WHAT IT SHOWS

The flagship view. Each card pairs a risk-classified conjunction with a recommended action: HIGH/MEDIUM/LOW band, Pc, miss distance, time to TCA, suggested ΔV , and cascade screening result.

HOW TO USE

- Review situation assessment text for each card
- Approve maneuver, continue monitoring, or define a custom burn
- Use Scan & Decide to refresh the full per-satellite sweep
- Funnel chart shows total alerts narrowed to critical

ENGINE BEHAVIOR

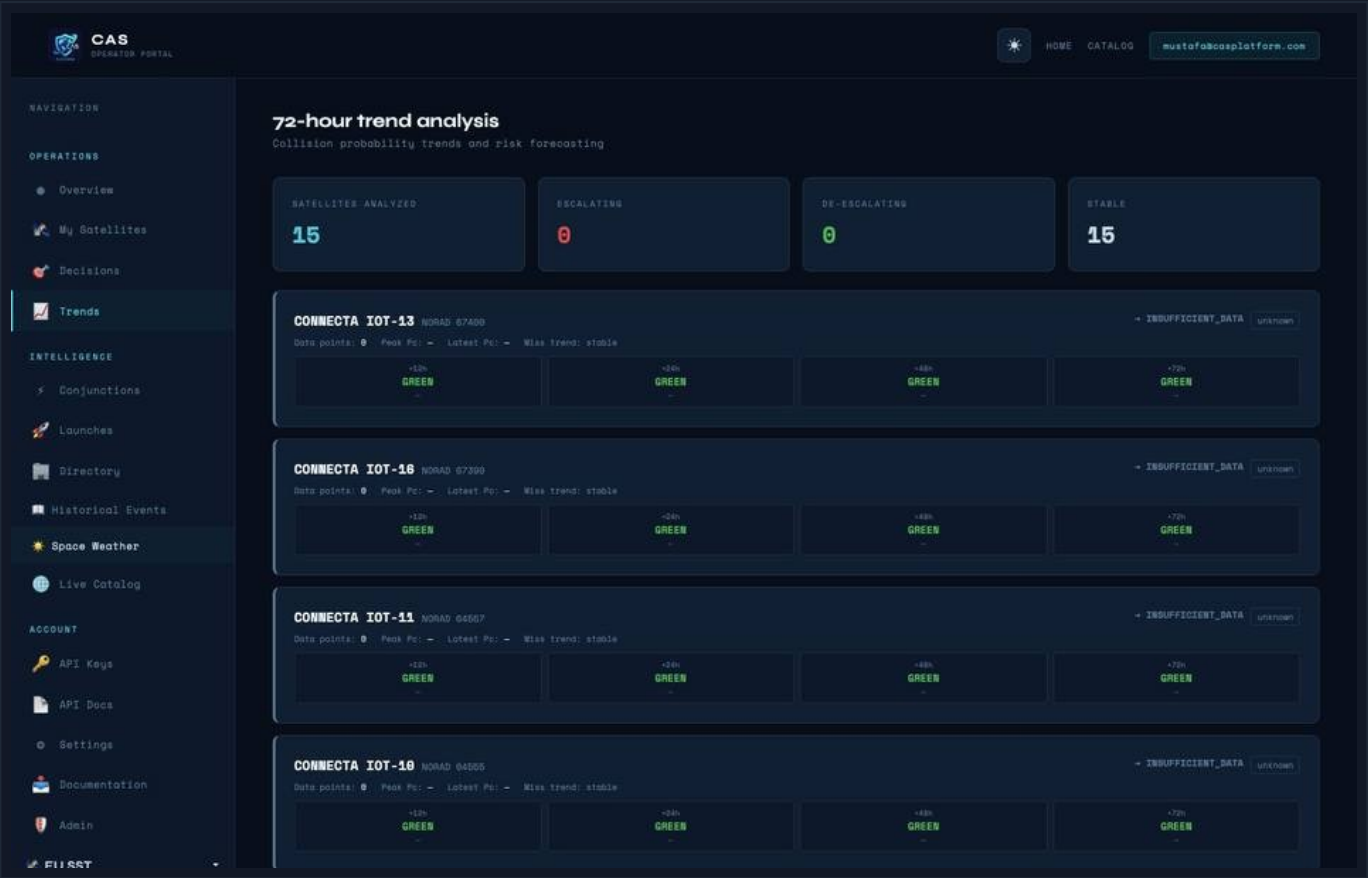
Decision Engine v0.7 + Cascade Analysis v2 screen the proposed maneuver against 28,510+ catalog objects to confirm no secondary conflicts. Foster/Chan Pc verified against analytical closed-form solution (10^{-3} tolerance).

OPERATIONAL VALUE

Translates raw probabilities into a clear operator action with quantified ΔV and fuel cost (~14–25 g for 150 kg sat class). Decision support — not autonomous control.

72-Hour Trend Analysis

Pc evolution per satellite – escalating, stable, de-escalating



WHAT IT SHOWS

Forward-looking risk view. For each watchlist satellite, Pc trajectory at +12h, +24h, +48h, +72h with classification (rising / stable / declining).

HOW TO USE

- Read the four headline counters: analyzed, escalating, de-escalating, stable
- Inspect each satellite card for current Pc and miss trend
- Confidence flag indicates data sufficiency

ENGINE BEHAVIOR

Engine snapshots Pc per CDM update and projects forward with conjunction-window decay models. Insufficient data flagged when fewer than three observations available.

OPERATIONAL VALUE

Buys lead time. Operators see escalating events earlier in the conjunction window, often before formal recommendation thresholds are crossed.

Live Conjunction Feed

Full Space-Track CDM history with risk banding



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FIISST

Conjunction events

Full history of tracked conjunction events

80 unique events (177 total records)

EXPORT CSV

RISK	OBJECT 1	OBJECT 2	MISS (M)	PC	TCA	CDM ID
RED	CZ-6A DEB	FENGYUN 1C DEB	6	1 in 175	2020-04-24T14:07	1A48881301
RED	FENGYUN 1C DEB	CZ-6A DEB	2	1 in 134	2020-04-24T14:07	1A48478991
RED	CZ-6A DEB	FENGYUN 1C DEB	4	1 in 167	2020-04-24T14:07	1A48518367
RED	FENGYUN 1C DEB	CZ-6A DEB	4	1 in 167	2020-04-24T14:07	1A49031870
RED	FENGYUN 1C DEB	CZ-6A DEB	4	1 in 123	2020-04-24T14:07	1A48907308
RED	CZ-6A DEB	FENGYUN 1C DEB	4	1 in 123	2020-04-24T14:07	1A48996809
RED	FENGYUN 1C DEB	CZ-6A DEB	10	1 in 119	2020-04-24T14:07	1A50762126
RED	FENGYUN 1C DEB	CZ-6A DEB	3	1 in 110	2020-04-24T14:07	1A51002631
RED	FENGYUN 1C DEB	METEOR 1-10	78	1 in 134	2020-04-23T15:00	1A47856832
RED	FENGYUN 1C DEB	METEOR 1-10	37	2.30%	2020-04-23T15:00	1A48476162
RED	METEOR 1-10	FENGYUN 1C DEB	37	2.30%	2020-04-23T15:00	1A48486204
RED	FENGYUN 1C DEB	METEOR 1-10	63	1.14%	2020-04-23T15:00	1A49031370
RED	METEOR 1-10	FENGYUN 1C DEB	63	1.14%	2020-04-23T15:00	1A48960746
RED	METEOR 1-10	FENGYUN 1C DEB	73	1 in 141	2020-04-23T15:00	1A48147948
RED	DMSP 50-2 F13 DEB	OV1-20/OV1-21 AKM	39	1 in 194	2020-04-23T00:11	1A48873364
RED	DMSP 50-2 F13 DEB	OV1-20/OV1-21 AKM	27	1 in 160	2020-04-23T00:11	1A47369338
RED	OV1-20/OV1-21 AKM	DMSP 50-2 F13 DEB	27	1 in 160	2020-04-23T00:11	1A47209325

WHAT IT SHOWS

The complete tracked-event ledger. Each row carries risk band, both objects, miss distance, Pc, TCA and the originating CDM ID. CSV export for offline analysis.

HOW TO USE

- Review unique-events count vs total records
- Sort by Pc or miss distance to triage
- Click a row for full CDM-level context
- Export CSV to share with mission planners or auditors

ENGINE BEHAVIOR

Hourly Space-Track sync with UPSERT on cdm_id — events update in place rather than duplicating. Foster/Chan Pc computed at ingest; risk band assigned by configurable thresholds.

OPERATIONAL VALUE

Single source of truth for what was reported, when, and how it was scored. Auditable and exportable for regulatory or insurance-grade documentation.

Launch Schedule

Upcoming and recent global launches

UPCOMING

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EXIT

Launch Schedule

Real-time launch data from TheSpaceDevs

UPCOMING

RECENT

Long March 6 | Unknown Payload

China Aerospace Science and Technology Corporation | Unknown

Details TBD

Taiyuan Satellite Launch Center, People's Republic of China

2025-04-28 12:00 UTC

T-4h 59m

Bojue 2.1a | Progress MS-34 (ISS)

Roscosmos Federal Space Agency (Roscosmos) | Roscosmos

Progress resupply mission to the International Space Station

Saturn Launch Complex, Republic of Kazakhstan

2025-04-28 01:00 UTC

T-15h 0m

Bojue-5 | Demo Flight

WU Spaceport | Test Flight

Demonstration flight for Russia's new Bojue-5 launch vehicle, with a mass simulator on board. Details TBD.

Saturn Launch Complex, Republic of Kazakhstan

2025-04-28 12:00 UTC

T-1d 3h

Falcon 9 Block 5 | Starlink Group 17-16

SpaceX | Communications

A batch of 58 satellites for the Starlink mega-constellation - SpaceX's project for space-based internet communication system.

Van der Bilt, CA, USA

2025-04-28 01:00 UTC

T-1d 6h

Falcon Heavy | ViaSat-3 F3 (ViaSat-3 Asia-Pacific)

SpaceX | Communications

The ViaSat-3 is a series of three Ka-band satellites is expected to provide vastly superior capabilities in terms of service speed and flexibility for a satellite platform. Each ViaSat-3 class satelli...

Kennedy Space Center, FL, USA

2025-04-27 14:00 UTC

T-2d 7h

Atlas V 551 | Amazon Lex (LA-96)

2025-04-27 14:00 UTC

T-2d 17h

RECENT

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Launch Schedule

Real-time launch data from TheSpaceDevs

UPCOMING

RECENT

Long March 2D | 4 x Sentinel test satellites

China Aerospace Science and Technology Corporation | Communications

Officially described as "Satellite-Internet Technology Demonstration Satellite". Probably 4 test satellites for the Chinese state-owned life communication satellite constellation Beidou.

Taiyuan Satellite Launch Center, People's Republic of China

2025-04-28 01:00 UTC

Launched

Angara 5.1 | Kosmos (Unknown Payload)

Roscosmos Space Forces | Government/Top Secret

Unknown satellite(s) for the Russian military.

Plesetsk Cosmodrome, Russian Federation

2025-04-28 01:00 UTC

Launched

Falcon 9 Block 5 | Starlink Group 17-14

SpaceX | Communications

A batch of 58 satellites for the Starlink mega-constellation - SpaceX's project for space-based internet communication system.

Van der Bilt, CA, USA

2025-04-28 01:00 UTC

Launched

Electron | Kakushin Rising (JAXA Rideshare)

Rocket Lab | Technology

JAXA-manifested rideshare of eight separate spacecraft that includes educational small sats, an ocean monitoring satellite, a demonstration satellite for ultra-small multi-spectral sensors, and a dog!

Rocket Lab Launch Complex 1, Mahia Peninsula, New Zealand

2025-04-28 01:00 UTC

Launched

HASTE | Bubbles

Rocket Lab | Government/Top Secret

Sub-orbital launch under Rocket Lab's Hyperbolic Accelerator Suborbital Test Electron (HASTE) program. Details TBD.

Wallops Flight Facility, Virginia, USA

2025-04-28 01:00 UTC

Launched

WHAT IT SHOWS

Real-time launch schedule from TheSpaceDevs. Upcoming countdowns, recent results, payload classification and launch site.

HOW TO USE

- Switch between Upcoming and Recent tabs
- Tap Go to open launch detail
- Scan payload class for relevance to your orbits
- Track new constellation deployments early

ENGINE BEHAVIOR

Polled from TheSpaceDevs API. Status, T-minus and outcome update automatically; results feed the catalog ingest pipeline.

OPERATIONAL VALUE

Anticipates near-term population growth in your orbital regime. Helps planners allocate review attention before new objects appear in CDMs.

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Space Industry Directory

Operators, agencies, manufacturers, SSA & launch providers

The screenshot displays the 'Space Industry Directory' interface within the CAS Operator Portal. The left sidebar contains navigation links: Overview, My Satellites, Decisions, Trends, Conjunctions, Launches, Directory (selected), Historical Events, Space Weather, Live Catalog, API Keys, API Docs, Settings, Documentation, Admin, and FIISST. The main content area is titled 'Space Industry Directory' and includes a search bar and filters for ALL, OPERATORS, AGENCIES, MANUFACTURERS, SSA, and LAUNCH. A summary bar shows 'Total: 2351 entries - Operators: 2328 Agencies: 11 SSA: 8 Manufacturers: 4 Launch: 3'. The directory grid lists the following entries:

Organization	Category	Location	Constellation	Satellites	Description	Website	Contact
SpaceX	OPERATOR	United States - Hawthorne, California	Starlink	1667	Designs, manufactures and launches rockets and spacecraft. Operates Starlink broadband constellation.	Website	Contact
Eutelsat Group	OPERATOR	France - Issy-les-Moulineaux, France	Eutelsat/OneWeb	713	European satellite telecommunications operator. Merged with OneWeb in 2023.	Website	Contact
OneWeb	OPERATOR	United Kingdom - London, UK	OneWeb	408	Global communications company building a LEO satellite constellation for broadband connectivity.	Website	Contact
NASA	AGENCY	United States - Washington, DC		248	National Aeronautics and Space Administration.	Website	Contact
SES S.A.	OPERATOR	Luxembourg - Betzdorf, Luxembourg	QSO mPOWER	239	Global satellite operator providing video and data connectivity worldwide.	Website	OTC:SESAP
US Space Force	AGENCY	United States - Arlington, Virginia		220	United States Space Force - military space operations.	Website	
Shanghai Spacecom (GeeSpace)	OPERATOR	China - Shanghai, China	Qianfan/G60	188	Chinese LEO broadband constellation operator (G60/Qianfan).	Website	Contact
Roscosmos	AGENCY	Russia - Moscow, Russia		177	Russian Federal Space Agency.	Website	
European Space Agency	AGENCY	Europe - Paris, France		113	Intergovernmental organisation dedicated to space exploration and research.	Website	Contact
ISRO	AGENCY						
JAXA	AGENCY						
CHSA	AGENCY						

WHAT IT SHOWS

Searchable registry of 2,351 entries across operators, agencies, manufacturers, SSA providers and launch organisations — with constellation, satellite count and contact info.

HOW TO USE

- Filter by category tab (Operators, Agencies, Manufacturers, SSA, Launch)
- Search by name or constellation
- Use Contact links to reach counterparts directly

ENGINE BEHAVIOR

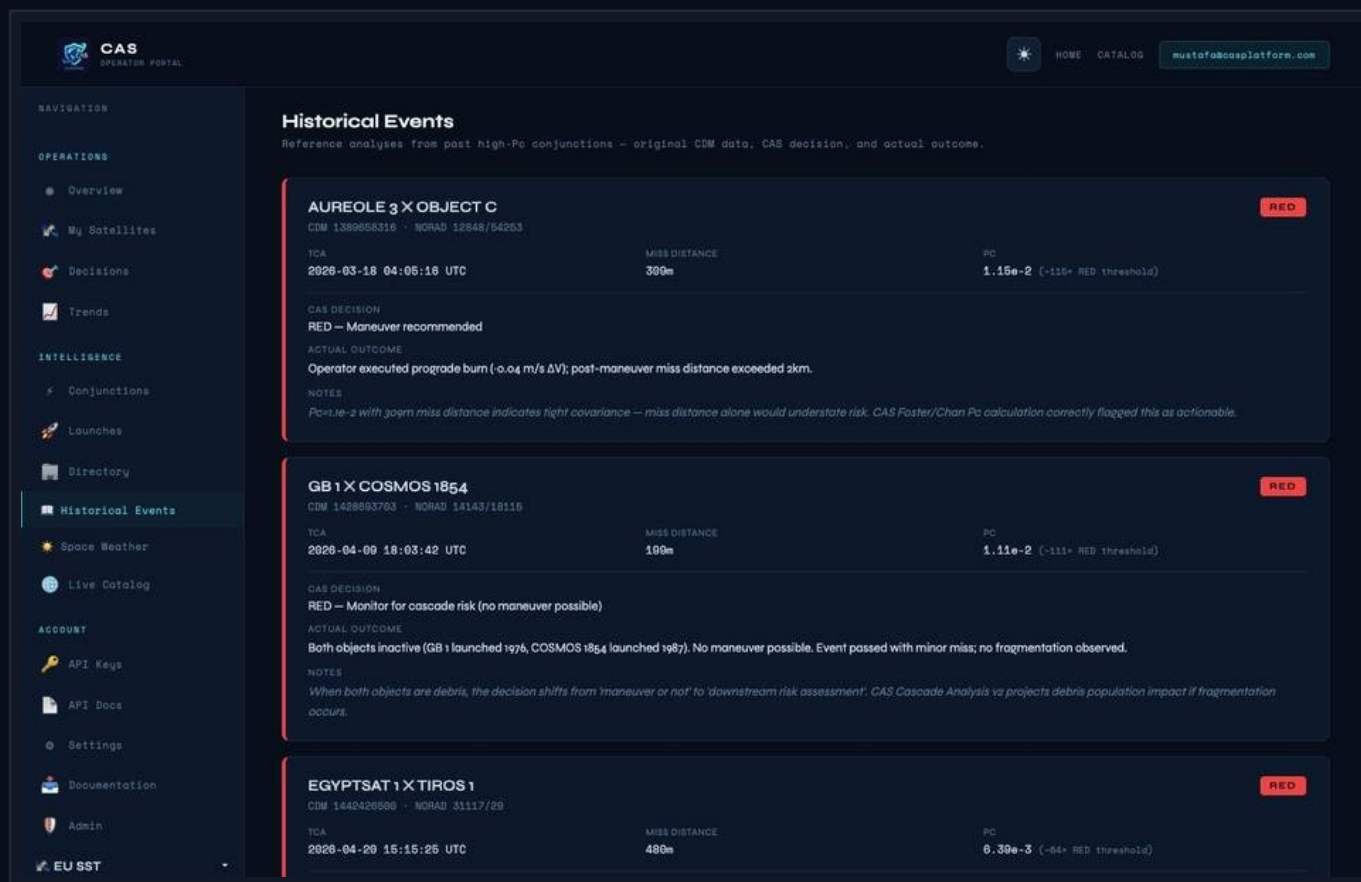
Curated registry with manual and feed-driven updates. Cross-linked to NORAD catalog so an operator's satellite count reflects active payloads.

OPERATIONAL VALUE

When a CDM names an unfamiliar counterpart, the directory closes the gap fast — operator, fleet size, contact channel — speeding coordinated maneuver decisions.

Reference High-Pc Events

Original CDM data, CAS decision and actual outcome



Historical Events
Reference analyses from past high-Pc conjunctions — original CDM data, CAS decision, and actual outcome.

Event	CDM ID	NORAD ID	TCA	MISS DISTANCE	PC	CAS DECISION	ACTUAL OUTCOME	NOTES
AUREOLE 3 X OBJECT C	1389658316	12848/64263	2026-03-18 04:05:16 UTC	309m	1.15e-2 (-110+ RED threshold)	RED — Maneuver recommended	Operator executed prograde burn (-0.04 m/s ΔV); post-maneuver miss distance exceeded 2km.	Pc=1.1e-2 with 309m miss distance indicates right covariance — miss distance alone would understate risk. CAS Foster/Chan Pc calculation correctly flagged this as actionable.
GB 1 X COSMOS 1854	1428663763	14143/18116	2026-04-09 18:03:42 UTC	190m	1.11e-2 (-111+ RED threshold)	RED — Monitor for cascade risk (no maneuver possible)	Both objects inactive (GB 1 launched 1976, COSMOS 1854 launched 1987). No maneuver possible. Event passed with minor miss; no fragmentation observed.	When both objects are debris, the decision shifts from 'maneuver or not' to 'downstream risk assessment'. CAS Cascade Analysis v2 projects debris population impact if fragmentation occurs.
EGYPTSAT 1 X TIROS 1	1442426509	31117/29	2026-04-20 15:15:26 UTC	480m	6.39e-3 (-64+ RED threshold)			

WHAT IT SHOWS

Six reference high-Pc conjunctions covering different regimes — covariance-driven, debris-debris, heritage assets, active constellations, emerging debris — each with the original CDM, CAS recommendation, and the actual operator outcome.

HOW TO USE

- Open a card to read the full case context
- Compare Pc, miss distance and CAS verdict
- Review notes for analyst commentary
- Use as training material for new mission planners

ENGINE BEHAVIOR

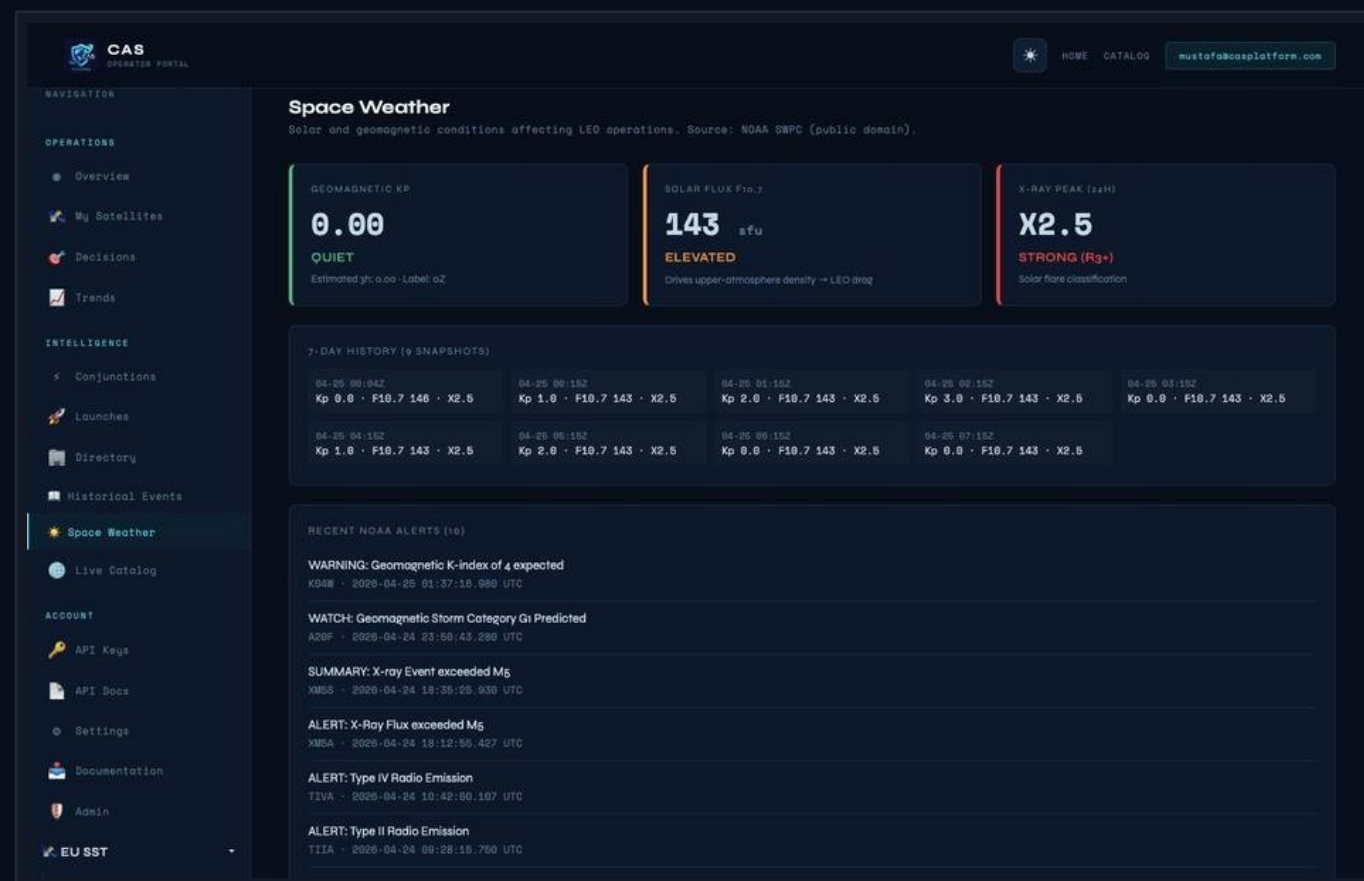
Static reference set drawn from real Space-Track CDMs. Pc values shown in scientific notation alongside threshold context (e.g. 1.15e-2, ~115× RED threshold). No synthetic data.

OPERATIONAL VALUE

Provides ground truth on how the engine reasons about edge cases — covariance squeeze, downstream cascade risk, debris-vs-debris. Builds operator confidence in the decision layer.

Space Weather Conditions

Geomagnetic Kp, F10.7 solar flux, X-ray peak – from NOAA SWPC



The screenshot shows the 'Space Weather' section of the CAS Operator Portal. The interface includes a sidebar with navigation links: Overview, My Satellites, Decisions, Trends, Conjunctions, Launches, Directory, Historical Events, Space Weather (selected), Live Catalog, API Keys, API Docs, Settings, Documentation, and Admin. The main content area displays three primary indicators: Geomagnetic Kp (0.00, QUIET), Solar Flux F10.7 (143 sfu, ELEVATED), and X-Ray Peak (X2.5, STRONG (R3+)). Below these are two sections: '7-DAY HISTORY (9 Snapshots)' showing a grid of Kp, F10.7, and X2.5 values for various dates, and 'RECENT NOAA ALERTS (16)' listing warnings, watches, and alerts related to geomagnetic storms and X-ray events.

WHAT IT SHOWS

Three primary indices for LEO operations: geomagnetic Kp (storm scale), F10.7 solar flux (drives upper-atmosphere density), X-ray peak (flare classification). 7-day history and recent NOAA alerts.

HOW TO USE

- Read the three headline cards for current state
- Inspect the snapshot grid for the last 7 days
- Scan recent NOAA alerts for warning context
- Factor $Kp \geq 5$ into TLE accuracy assumptions before maneuvering

ENGINE BEHAVIOR

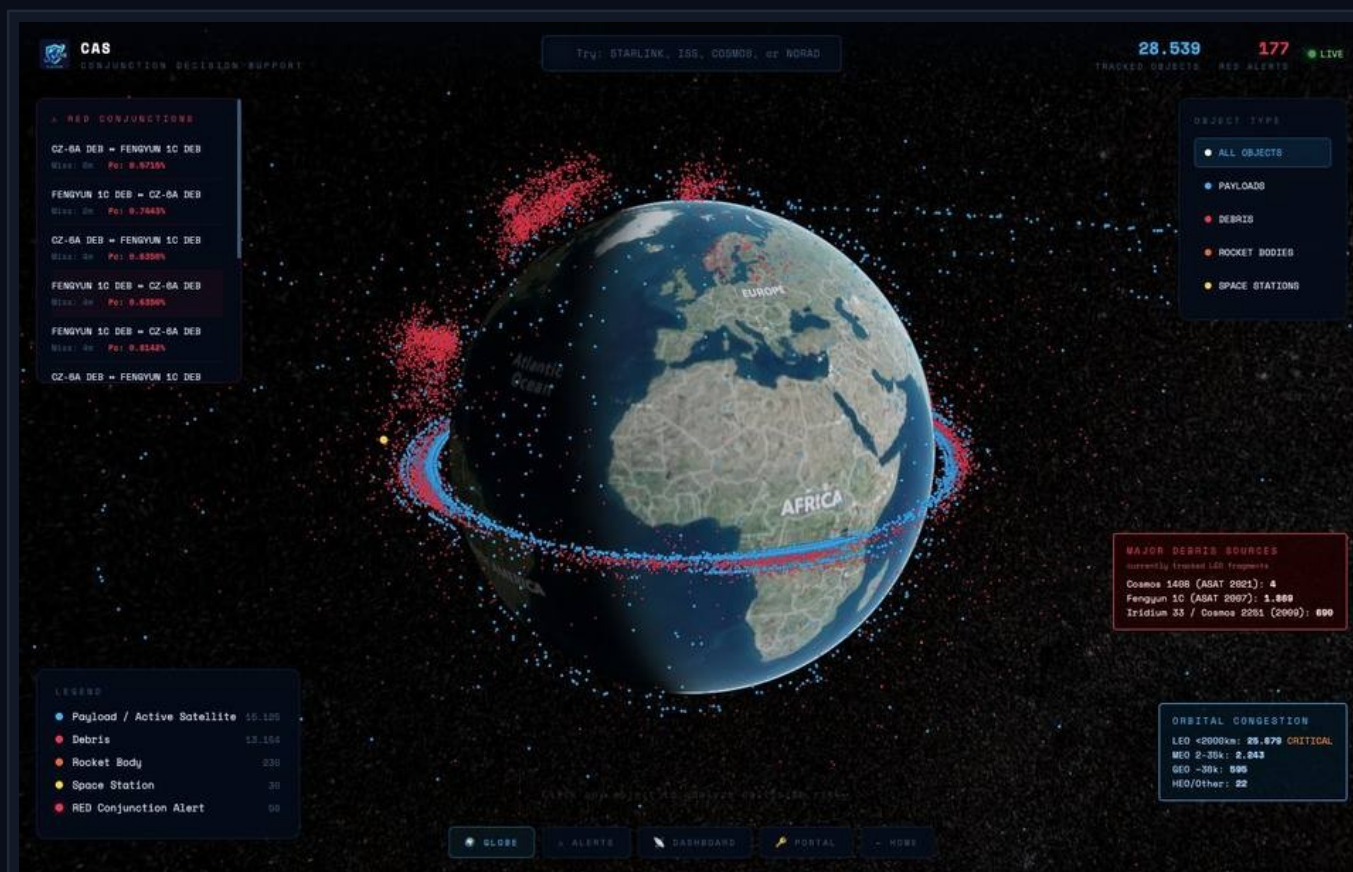
Hourly sync from NOAA SWPC public feeds. Indices stored alongside conjunction data so retrospective analysis can correlate Pc shifts with geomagnetic events.

OPERATIONAL VALUE

Geomagnetic storms inflate atmospheric drag, drift TLEs and shift conjunction geometry. This panel keeps that variable visible without leaving the portal.

Live 3D Catalog

28,510+ tracked objects, multi-source ingestion



WHAT IT SHOWS

3D CesiumJS globe rendering the full tracked catalog — payloads, debris, rocket bodies, space stations — with real-time RED conjunctions, orbital congestion summary and major debris sources.

HOW TO USE

- Filter by object type using the right-side legend
- Click any object to inspect its conjunction risk
- Inspect Major Debris Sources to size historical events
- Use Orbital Congestion to gauge regime saturation

ENGINE BEHAVIOR

Multi-source ingestion: Celestrak (9 groups) plus Space-Track LEO debris and rocket bodies. 6-hour cache, dedup on NORAD ID, full TLE refresh into the engine.

OPERATIONAL VALUE

Public SSA pictures often show only large objects; the CAS catalog matches the conjunction-screening engine. What you see here is what the engine actually reasons over.

Programmatic Access

REST API – keys, endpoints and examples

API KEYS

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API keys

Manage your API authentication credentials

YOUR API KEY

Use this key for programmatic access to the CAS API.

cas_4731691962238c4c93654a8998983473

COPY

REGENERATE KEY

USAGE EXAMPLES

Using API Key as query parameter
curl "https://www.casplatform.com/api/history?limit=10&api_key=cas_4731691962238c4c93654a8998983473"

Using API Key in header
curl -H "Authorization: ApiKey cas_4731691962238c4c93654a8998983473" \\\n"https://www.casplatform.com/api/history?limit=10"

Login to get JWT token
curl -X POST https://www.casplatform.com/api/auth/login \\\n-H "Content-Type: application/json" \\\n-d '{"email":"you@email.com","password":"yourpass"}'

Use JWT token
curl -H "Authorization: Bearer YOUR_TOKEN" \\\n"https://www.casplatform.com/api/auth/me"

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ACCOUNT

API Keys

API Docs

Settings

Documentation

Admin

Full Screen

API documentation

Complete REST API reference for CAS Platform v0.9

BASE URL: <https://www.casplatform.com/api/> | AUTH: [Bearer Token](#) or [ApiKey](#) header | FORMAT: [JSON](#)

GET /api/health

Display health check

GET /api/status

API status dashboard

GET /api/metrics

Prometheus-compatible metrics

GET /api/history?limit=N

Derivative event history

POST /api/analyze

Analyze TLE data for conjunctions

POST /api/spacetrack

Fetch CDM form Space-Track

POST /api/auth/register

Create new operator account

POST /api/auth/login

Login and get JWT token

GET /api/auth/me

Get current user info (requires auth)

WHAT IT SHOWS

Per-account API key plus a complete REST reference — 40+ endpoints covering health, decisions, history, analyze, Space-Track and auth.

HOW TO USE

- Copy your cas_xxx key — keep it secret
- Authenticate via ApiKey header or Bearer JWT
- Reference /api/health for status checks
- Regenerate the key if it is compromised

ENGINE BEHAVIOR

Key auth maps directly to user identity and tier. JWT alternative for short-lived sessions. JSON responses everywhere; rate limits on public endpoints.

OPERATIONAL VALUE

Drop CAS decisions into existing flight-dynamics pipelines, dashboards or alerting systems without leaving your stack.

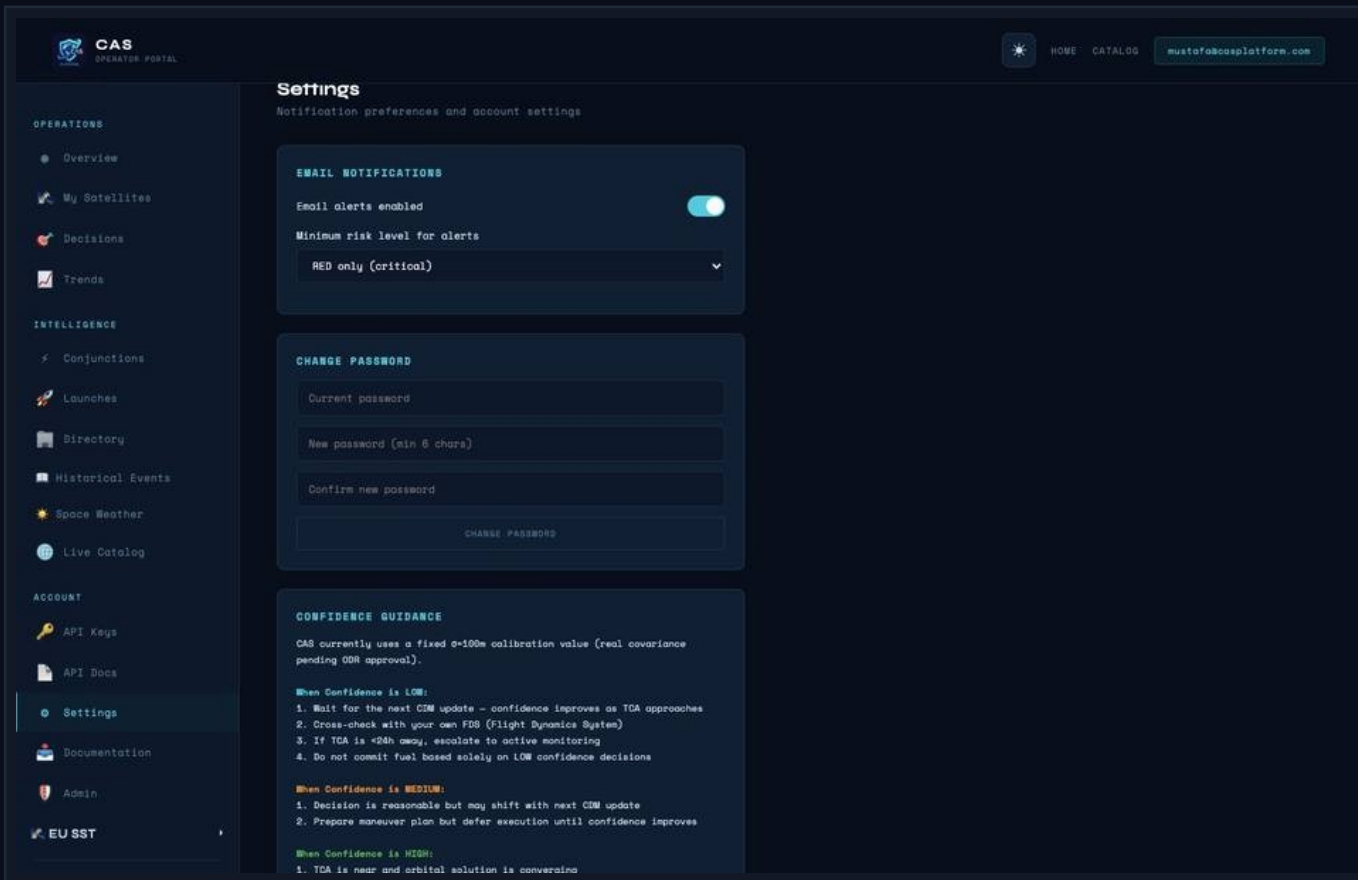
casplatform.com

CAS Platform – Operator Portal User Guide

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Notifications & Preferences

Email alerts, password, confidence guidance



The screenshot shows the 'Settings' page of the CAS Operator Portal. The page is divided into three main sections: 'EMAIL NOTIFICATIONS', 'CHANGE PASSWORD', and 'CONFIDENCE GUIDANCE'. The left sidebar contains navigation links for 'OPERATIONS' (Overview, My Satellites, Decisions, Trends), 'INTELLIGENCE' (Conjunctions, Launches, Directory, Historical Events, Space Weather, Live Catalog), 'ACCOUNT' (API Keys, API Docs, Settings, Documentation, Admin), and 'EU SST'. The top right of the page shows a user profile for 'mustafab@casplatform.com'.

Settings
Notification preferences and account settings

EMAIL NOTIFICATIONS

Email alerts enabled ☒

Minimum risk level for alerts
RED only (critical)

CHANGE PASSWORD

Current password

New password (min 6 chars)

Confirm new password

CHANGE PASSWORD

CONFIDENCE GUIDANCE

CAS currently uses a fixed $\sigma=100m$ calibration value (real covariance pending ODR approval).

When Confidence is LOW:

1. Wait for the next CDM update – confidence improves as TCA approaches
2. Cross-check with your own FDS (Flight Dynamics System)
3. If TCA is $<24h$ away, escalate to active monitoring
4. Do not commit fuel based solely on LOW confidence decisions

When Confidence is MEDIUM:

1. Decision is reasonable but may shift with next CDM update
2. Prepare maneuver plan but defer execution until confidence improves

When Confidence is HIGH:

1. TCA is near and orbital solution is converging

WHAT IT SHOWS

Account-level controls: enable email alerts, set the minimum risk threshold for notifications, change password, and read confidence-level operational guidance.

HOW TO USE

- Toggle email alerts on or off
- Pick minimum severity (RED only by default)
- Update password — minimum 6 characters
- Read the LOW / MEDIUM / HIGH confidence playbook

ENGINE BEHAVIOR

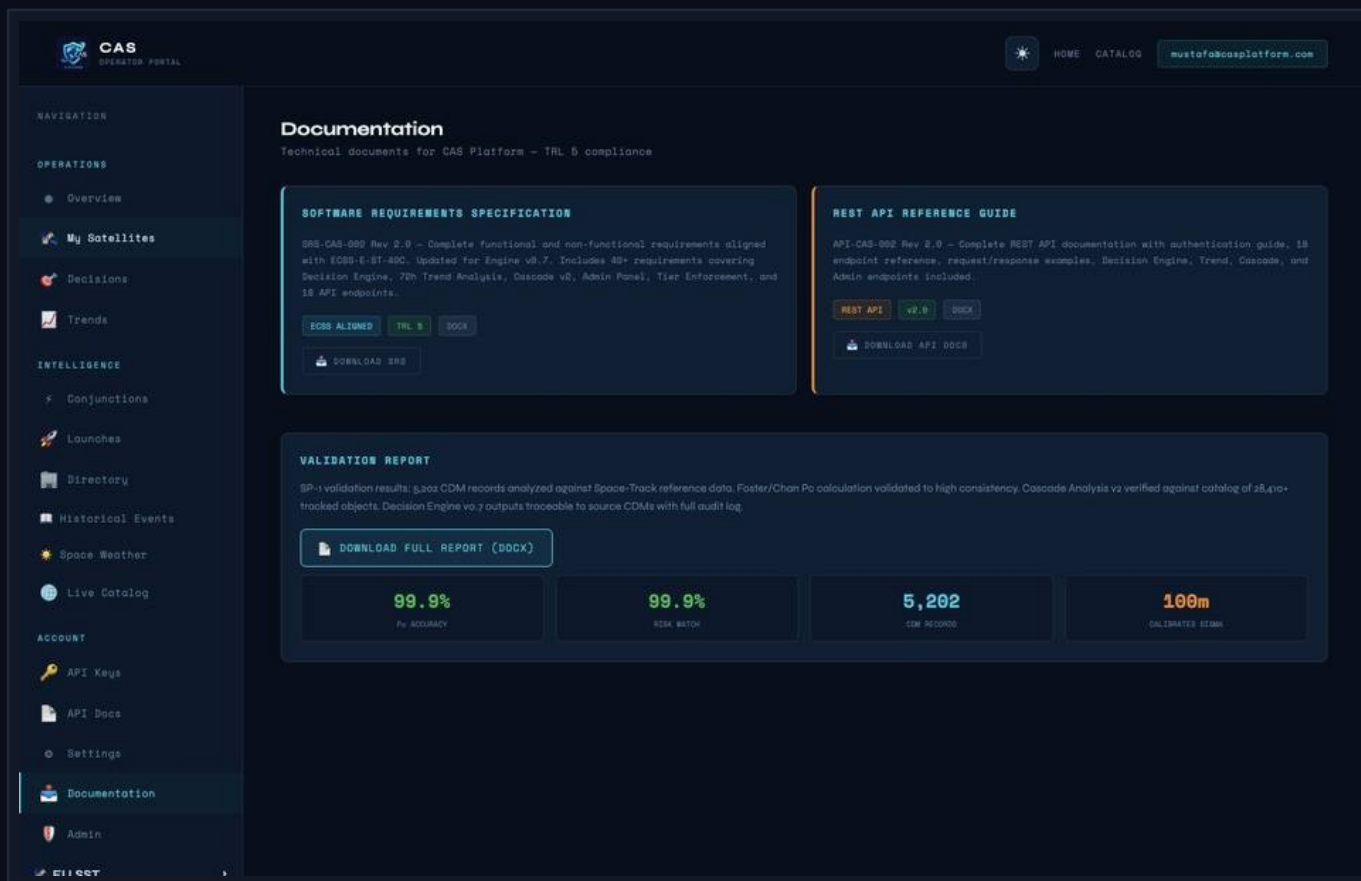
Confidence guidance is procedural: it tells operators how to handle decisions while real covariance calibration ($\sigma=100$ m placeholder) is pending ODR approval.

OPERATIONAL VALUE

Filters alert volume to operationally meaningful events and gives an explicit playbook for acting on the engine's confidence band.

Technical Documentation

SRS, REST API reference, validation report



The screenshot shows the CAS Operator Portal interface. On the left is a dark sidebar with a navigation menu. The main content area is titled 'Documentation' and contains three sections: 'SOFTWARE REQUIREMENTS SPECIFICATION', 'REST API REFERENCE GUIDE', and 'VALIDATION REPORT'. Each section has a brief description and a 'DOWNLOAD' button. The 'VALIDATION REPORT' section also displays four key performance indicators (KPIs) in a row.

NAVIGATION

OPERATIONS

- Overview
- My Satellites
- Decisions
- Trends

INTELLIGENCE

- Conjunctions
- Launches
- Directory
- Historical Events
- Space Weather
- Live Catalog

ACCOUNT

- API Keys
- API Docs
- Settings
- Documentation
- Admin

Documentation

Technical documents for CAS Platform – TRL 5 compliance

SOFTWARE REQUIREMENTS SPECIFICATION

SRS-CAS-002 Rev 2.0 – Complete functional and non-functional requirements aligned with ECSS-E-ST-40C. Updated for Engine v0.7. Includes 40+ requirements covering Decision Engine, 70h Trend Analysis, Cascade v0, Admin Panel, Tier Enforcement, and 18 API endpoints.

ECSS ALIGNED TRL 5 DOCX

DOWNLOAD SRS

REST API REFERENCE GUIDE

API-CAS-002 Rev 2.0 – Complete REST API documentation with authentication guide, 18 endpoint reference, request/response examples, Decision Engine, Trend, Cascade, and Admin endpoints included.

REST API v0.9 QUICK

DOWNLOAD API DOCS

VALIDATION REPORT

SP-1 validation results: 5,202 CDM records analyzed against Space-Track reference data. Foster/Chan Pc calculation validated to high consistency. Cascade Analysis v2 verified against catalog of 28,410+ tracked objects. Decision Engine v0.7 outputs traceable to source CDMs with full audit log.

DOWNLOAD FULL REPORT (DOCX)

99.9% Pc ACCURACY	99.9% RISK RATION	5,202 CDM RECORDS	100m ORIGINALIZED CDMs
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WHAT IT SHOWS

Three downloadable artefacts: Software Requirements Specification (ECSS-aligned), full REST API reference, and the SP-1 Validation Report with audit-grade backing.

HOW TO USE

- Download SRS for procurement and integration teams
- Download API docs for developer onboarding
- Download the full Validation Report (DOCX) for technical review
- Cite version stamps in your safety case

ENGINE BEHAVIOR

Validation: 84 automated unit tests, 100% pass rate; Foster/Chan Pc verified against analytical closed-form (10^{-3} tolerance); 5,202 CDM validation reference set; Cascade Analysis v2 verified across 28,510+ tracked objects with full audit log.

OPERATIONAL VALUE

Gives operators, regulators and insurers an evidence trail. The decision layer's behaviour is documented, version-controlled and reviewable — not a black box.

Get in touch

Independent decision support layer for EU LEO operators.



EMAIL

mustafa@casplatform.com

Direct line to the founder. Response within 24 hours.

WEB

casplatform.com

Pilot programme open for qualifying LEO operators.